

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination

May 2012

CL 102.01 Environmental Sciences

Date: 09.05.2012, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Discuss relationship between Man and Environment. [03]
(b) Write a short note on Biosphere. [04]
- Q - 2 (a) What are the causes of Environmental Pollution? List all Environmental Pollution and their sources in detail. [04]
(b) Define Noise Pollution. Describe all effects of Noise pollution. [05]
(c) Explain objectives & guiding principles of Environmental Studies. [05]

OR

- Q - 2 (a) Give general classification of pollutants and explain classification of pollutants based on its origin. [04]
(b) Describe the term Environmental degradation and list out all possible types of Environmental pollution. [05]
(c) Discuss the temperature profile of atmosphere in detail. [05]
- Q - 3 (a) Explain effects of air pollution on human being. [04]
(b) Describe sources and effects of water pollutants. [05]
(c) Describe the main causes of depletion of ozone layer and briefly comment over control measures taken to prevent further ozone depletion at international level. [05]

OR

- Q - 3 (a) Give classification of water pollutants and describe organic water pollutants. [04]
(b) Enlist all current global environmental problems and discuss green house effect in details. [05]
(c) Define Eutrophication, its causes and effects. [05]

SECTION – II

- Q - 4 (a) Define Ecology. Explain role of components of ecosystem. [03]
(b) What is nitrogen fixation and denitrification? [02]

- (c) Describe food chain. [02]
- Q - 5 (a) Compare conventional and non-conventional energy resources. [04]
- (b) What do you mean by Biogeochemical cycle? Describe phosphorous cycle with a neat sketch. [05]
- (c) Explain food web with example. [05]

OR

- Q - 5 (a) Write advantages and disadvantages of solar energy. [04]
- (b) Define ecological pyramid? Enlist its type and explain pyramid of biomass. [05]
- (c) Explain Estuarine ecosystem? [05]
- Q - 6 (a) What are the problems due to overexploitation of energy resources? [04]
- (b) Explain in detail the role of decomposers and transformers. [05]
- (c) 1. Environmental lapse rate is _____ [05]
2. Total available air mass is _____ in nature. [05]
3. Ozone is formed by _____ reaction. [05]
4. Define 'Primary pollutant'. [05]
5. Define 'Qualitative pollutant'. [05]

OR

- Q - 6 (a) Discuss in detail conservation v/s destruction of natural resources [04]
- (b) What are the objectives to study ecology? [05]
- (c) 1. As per EPA Environment is defined as _____ [05]
2. Total quantity of available water in environment is _____ [05]
3. Inner core of lithosphere contains _____ [05]
4. Define 'Point source pollutant'. [05]
5. Define 'Biodegradable pollutant'. [05]

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